

SAT Math · Test X

1

★ Mark for Review

ABC

For the equation, b and c are constants. For any real number t , the point $\left(\frac{-2t+4}{3}, t\right)$ lies on the graph of the equation $bx + 3y = c$. What is the value of c ?

A. -2 B. 4 C. 6 D. 12

2

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ABC

The function q is defined by

$$q(x) = a|x - 9|^2 - 87|x - 9| + b,$$

where a and b are constants and $a > b > 87$. In the xy -plane, the graph of $y = q(x)$ contains the points $(639, h)$ and $(-621, k)$, where h and k are constants. What is the value of $9(-639)^{h-k} + 621(9)^{k-h}$?

3

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ABC

In the given equation, a and b are positive constants. A solution to the equation is $x = 42$.

$$x(9x - 2a) + b(9x - 2a) - (x + b) = 0$$

What is the value of a ?

4

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ABC

The function z is defined by

$$z(w) = (0.829)^{2w}.$$

The value of $z(w)$ decreases by $p\%$ for each increase by 1 in the value of w . Which of the following is closest to the value of p ?

A. 0.171

B. 0.829

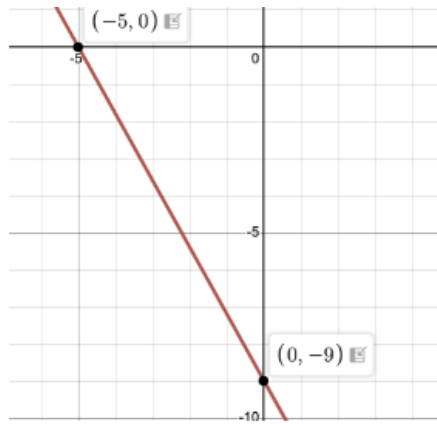
C. 17.1

D. 31.3

5

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ABC



The graph of line h is shown in the xy -plane. Line k (not shown) is defined by the equation $sx + 40y = t$, where s and t are constants. If line k is graphed in this xy -plane, the result is a graph of two linear equations. This system of two linear equations has no solution. Which of the following is NOT a possible value of t ?

A. -360 B. -9 C. 72 D. 200

6

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ABC

The function h is defined by

$$h(x) = -\sqrt{x^2 + bx + c},$$

where b and c are constants. The graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{358})$. If $h(m) = 0$, what is the greatest possible value of m ?

A. 2

B. 177

C. 179

D. 358

7

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ABC

The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$, and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

I. $f(x) = a(2.1)^{x+b}$

II. $g(x) = a(2.1)^x + b$

A. I only

B. II only

C. I and II

D. Neither I nor II

8

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ABC

The function f is defined by

$$f(x) = \frac{x^2 + ax + b}{2x + c},$$

where a , b , and c are constants. The graph of $y = f(x)$ does not intersect the line $x = 4$. If $f(5) = f(6) = 0$, what is the value of $a + b + c$?

A. 10**B.** 11**C.** 12**D.** 13

9

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ABC

The positive number a is 2,106% of the sum of the positive numbers b and c , and b is 81% of c . What percent of b is a ?

A. 4,706%**B.** 47.06%**C.** 470,600%**D.** 47,060%

10

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ABC

Point N lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point O is the center of the circle and has coordinates $(0, 0)$. Point M also lies on the unit circle, and the measure of angle $\angle NOM$ is $\frac{266\pi}{3}$ radians. If the coordinates of point M are (a, b) , where a and b are constants, what is the value of a ?

A. $\frac{1}{2}$

B. $\frac{\sqrt{3}}{2}$

C. $-\frac{1}{2}$

D. $-\frac{\sqrt{3}}{2}$

11

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ABC

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k(6a + b)$, where k is a constant. What is the value of k ?

$$24x^2 - (12a + 2b)x + ab = 0$$

A. $\frac{1}{12}$

B. $-\frac{1}{12}$

C. 12

D. -12

12

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ABC

In triangle ABC , the measure of angle B is 90° and BD is an altitude of the triangle. The length of $AB = 15$, and the length of AC is 23 greater than the length of AB . What is the value of $\frac{BC}{BD}$?

A. $\frac{15}{38}$

B. $\frac{15}{23}$

C. $\frac{23}{15}$

D. $\frac{38}{15}$

13

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ABC

A right square pyramid has a height of 11 centimeters (cm). A second right square pyramid has a height of 22 centimeters. The area of the bases of each of the two pyramids is 100 cm^2 . Which of the following is closest to the difference in the surface area of the second pyramid and the surface area of the first pyramid, in cm^2 ?

A. 209

B. 220

C. 367

D. 551

14

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ABC

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week:

- The number of prey had increased by 2,700% of the number of prey at the start of the week.
- The number of predators had increased by 180% of the number of predators at the start of the week.

The number of prey at the end of the week was $p\%$ greater than the number of predators at the end of the week. What is the value of p ?

A. 9

B. 90

C. 900

D. 9,000

15

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ABC

A beekeeper's initial observation of the population of a certain bee colony was 1,100 bees. The beekeeper set a goal to increase the population to 2,600 bees. The beekeeper uses a model that predicts the population of this bee colony:

- begins at 1,100
- increases by 120 bees per week in the first two weeks
- then increases by 180 bees per week until the goal is reached

According to this model, at the end of week w after the initial observation, where $w > 2$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

A. $p(w) = 2600 - 180w$

B. $p(w) = 2480 + 180w$

C. $p(w) = 1620 - 180w$

D. $p(w) = -120 + 180w$

16

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ABC

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

A. 0.3**B.** 16.3**C.** 195.8**D.** 220.4

17

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ABC

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

A. $3x^2 + 7x + 14b$

B. $3x^2 + 16x + 14b$

C. $3x^2 + 18x + 14b$

D. $3x^2 + 25x + 14b$

18

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ABC

In the given system of equations, t is a constant. If the system has no solution, what is the value of t ?

$$16x - 20y = 6y + 8$$

$$ty = 8x + \frac{1}{5}$$

19

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ABC

In the given equation, k is a positive constant. The product of the solutions to the equation is 72. What is the value of k ?

$$\frac{5}{9}(5x + 9)(x + \sqrt{5k + 9})(x - \sqrt{5k + 9}) = 0$$

20

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ABC

In the xy -plane, a unit circle with center at the origin O contains point A with coordinates $(1, 0)$ and point B with coordinates $\left(\frac{5}{\sqrt{34}}, -\frac{3}{\sqrt{34}}\right)$. If the measure of angle AOB is p radians, what is the value of $\frac{\cos p}{\sin p}$?

21

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ABC

A pyramid with square base area of 36 m^2 has a total surface area of $36 + 12\sqrt{493}\text{ m}^2$. Find the height of the pyramid.

22

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ABC

The height of a right circular cylinder is 36 inches, and the circumference of its base is 360 inches. Which expression represents the total surface area, in square inches, of the cylinder?

A. $(36)(360) + 2\pi \left(\frac{180}{\pi}\right)^2$

B. $(36)(360) + \pi \left(\frac{180}{\pi}\right)^2$

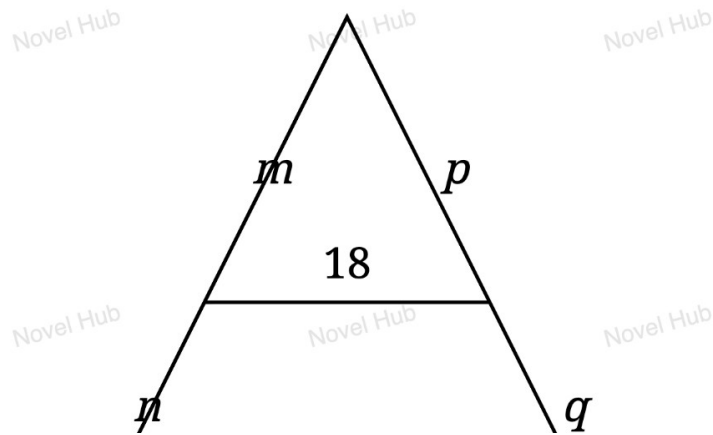
C. $\pi \left(\frac{180}{\pi}\right)^2 (36)$

D. $(36)(360)$

23

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ABC



The parallel line segment divides the other two sides of the triangle into segments of lengths m and n , and p and q , respectively. If $\frac{p}{q} = k$ and $\frac{m}{n} = k$, what is the value of k ?

24

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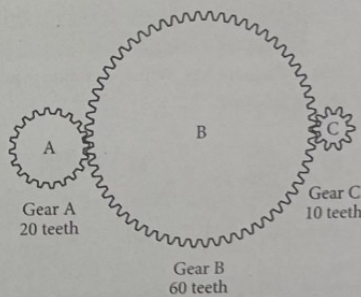
ABC

A gear ratio $r : s$ is the ratio of the number of teeth of two connected gears. The ratio of the number of revolutions per minute (rpm) of two gear wheels is $s : r$. In the diagram below, Gear A is turned by a motor. The turning of Gear A causes Gears B and C to turn as well.

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If Gear A is rotated by the motor at a rate of 100 rpm, what is the number of revolutions per minute for Gear C?

A) 50
B) 110
C) 200
D) 1,000



The diagram shows three gears labeled A, B, and C. Gear A is the smallest, Gear B is the largest, and Gear C is the smallest. Gear A is meshed with Gear B, and Gear B is meshed with Gear C. The number of teeth for each gear is indicated below it: Gear A has 20 teeth, Gear B has 60 teeth, and Gear C has 10 teeth.

If Gear A is rotated by the motor at a rate of 100 rpm, what is the number of revolutions per minute for Gear C?

A. 50**B.** 110**C.** 200**D.** 1,000